LP-EGR CONTROL VALVE [SKYACTIV-D 1.5]

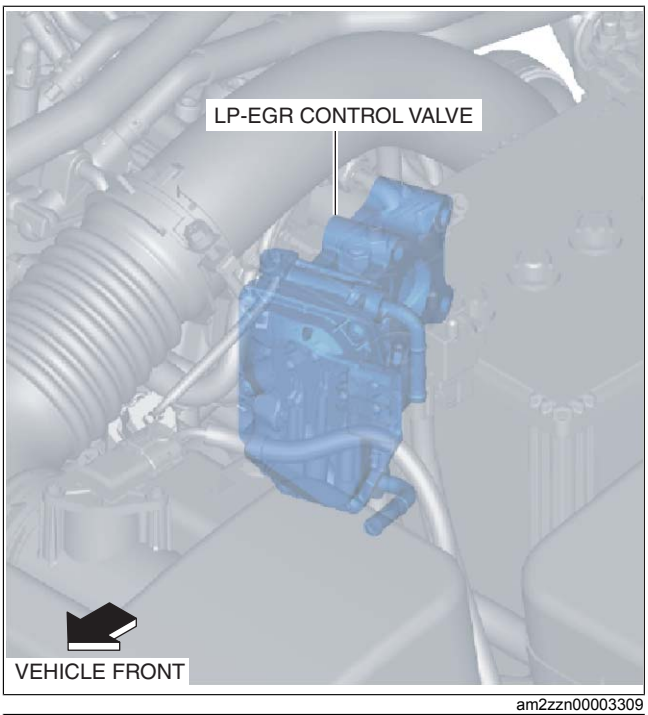
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Purpose, Function

- The LP-EGR control valve adjusts the exhaust gas volume recirculated to the variable geometry turbocharger according to the engine operation conditions to reduce the NOx generated largely when the combustion temperature in the combustion chamber is high.

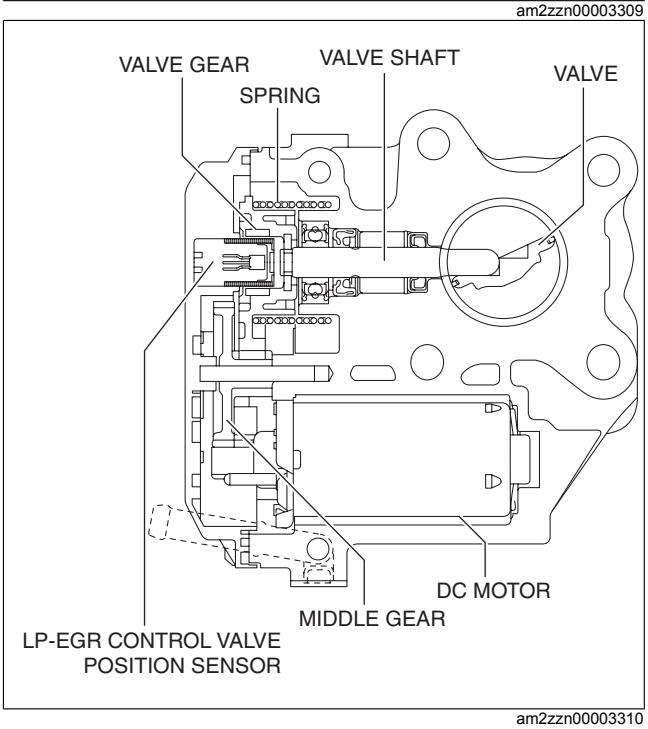
Construction

- The LP-EGR control valve is located at the rear of the engine compartment.



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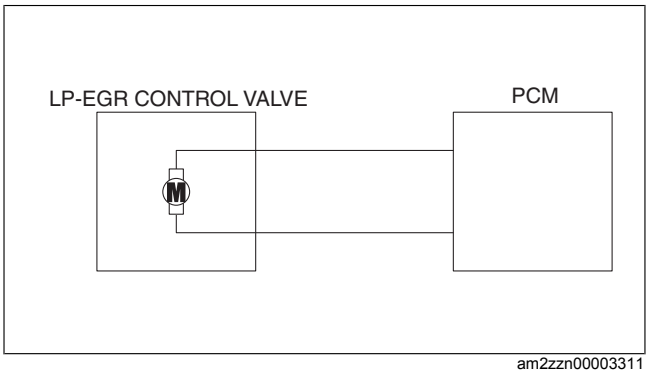
- The LP-EGR control valve consists of the DC motor, mid gear, valve gear, spring, valve shaft, and the valve.
- LP-EGR control valve has an integrated LP-EGR control valve position sensor. For details on the LP-EGR control valve position sensor, refer to the CONTROL SYSTEM. (See LP-EGR CONTROL VALVE POSITION SENSOR [SKYACTIV-D 1.5].)



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Operation

- The LP-EGR control valve is opened to the valve opening angle according to the energization current flowing to the DC motor from the PCM.
- The LP-EGR control valve performs coordinated control with the intake shutter valve, HP-EGR control valve, and exhaust shutter valve. For details on the LP-EGR control valve control, refer to the CONTROL SYSTEM. (See EGR CONTROL [SKYACTIV-D 1.5].)
- In the event that the LP-EGR control valve has a malfunction and the DC motor control is disabled, the shutter valve is closed to the fully closed position by the return spring. As a result, the air amount necessary to move the vehicle is assured.



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Fail-safe

DTC No.	Fail-safe function
P0407:00	• Limits the engine torque. • Inhibits the EGR control.
P0408:00	
P045B:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
P049C:00	
P213B:00	
P049B:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits engine-stop by operating the i-stop function.